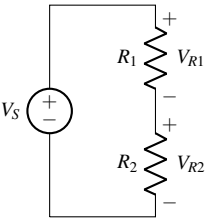
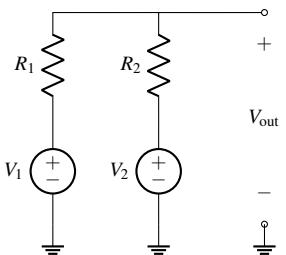
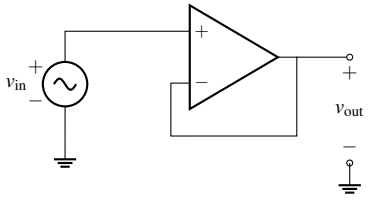
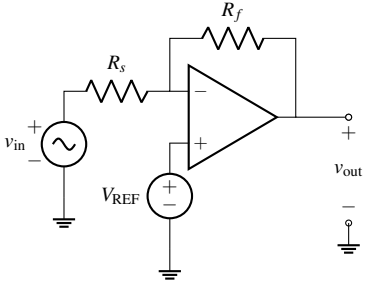
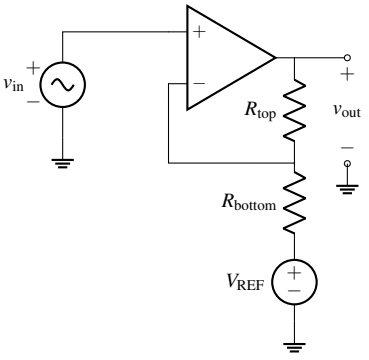


EECS 16A Designing Information Devices and Systems I

Discussion 11B

For Reference: Example Circuits

Voltage Divider  $V_{R2} = V_S \left(\frac{R_2}{R_1 + R_2} \right)$	Voltage Summer  $V_{out} = V_1 \left(\frac{R_2}{R_1 + R_2} \right) + V_2 \left(\frac{R_1}{R_1 + R_2} \right)$	Unity Gain Buffer  $\frac{v_{out}}{v_{in}} = 1$
Inverting Amplifier  $v_{out} = v_{in} \left(-\frac{R_f}{R_s} \right) + V_{REF} \left(\frac{R_f}{R_s} + 1 \right)$	Non-inverting Amplifier  $v_{out} = v_{in} \left(1 + \frac{R_{top}}{R_{bottom}} \right) - V_{REF} \left(\frac{R_{top}}{R_{bottom}} \right)$	

1. Noisy Parties (from Fa18 MT2)

To celebrate the (peaceful and very reasonable) end of the pizza-on-pineapple war, course staff decided to throw a party for the entire class! (Disclaimer: Not really) You and your friends want to take a picture to commemorate the night. In the imaging lab, you learned about how to make a camera with a single photodiode and a projector. Most modern cameras, however, have many photodiodes and do not need a sweeping light source like what you used in lab and need to work in non-ideal, non-laboratory conditions.

(a) We can model a photodiode as a capacitor in parallel with a current source:

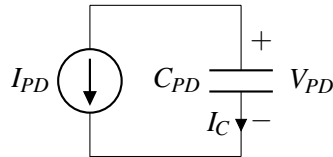


Figure 1: Photodiode circuit model

The current I_{PD} is related to the intensity of the light shining on the photodiode. Start with the following equations:

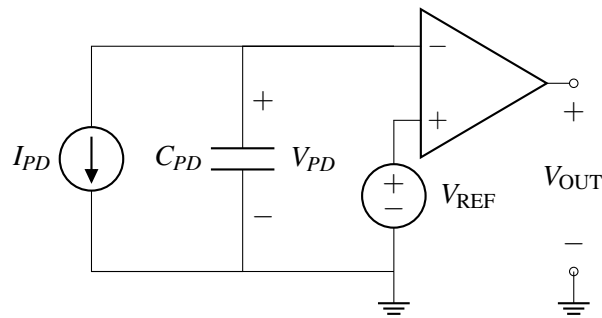
- $V_{PD}(t = 0) = V_{DD}$
- $I_C = C_{PD} \frac{dV_{PD}}{dt}$

Begin with the above equations and show that $V_{PD}(t) = V_{DD} - \frac{I_{PD}}{C_{PD}}t$.

Hint: Follow passive sign convention.

- (b) When feeding information from circuits into computers, we need to convert from our continuous-valued analog signals into discrete-valued digital signals the computer can understand using an analog-to-digital converter (ADC).

One type of ADC uses the time it takes for an event to occur in its conversion (converting time to the digital domain involves counting at a steady rate until the event occurs). We care about the time it takes for the comparator output to change.



Parameter	Unit	Value
I_{PD}	A	$1 \cdot 10^{-3}$
C_{PD}	F	$1 \cdot 10^{-9}$
V_{REF}	V	0.5
V_{DD}	V	1.0

Figure 2: Integrating ADC

Table 1: Numerical values for parts (b) and (c)

Draw $V_{PD}(t)$ in the graph below, where $V_{PD}(t = 0) = V_{DD}$. Clearly label the following with numerical values:

- Slope of the $V_{PD}(t)$ vs. t curve, i.e. $\frac{dV_{PD}}{dt}$.
- t_{flip} where $V_{PD}(t_{\text{flip}}) = V_{REF}$ and the comparator output goes high for the first time
- t_{zero} where $V_{PD}(t_{\text{zero}}) = 0V$

Hint: $1\mu\text{s} = 1 \cdot 10^{-6}\text{s}$.

- (c) Everyone in your picture is holding still, and in previous parts we assumed a constant current across the capacitor. However, someone just turned on strobe lights! You only want a picture of people, so we can treat the flashing lights as interference. Now our full circuit looks like Figure 3

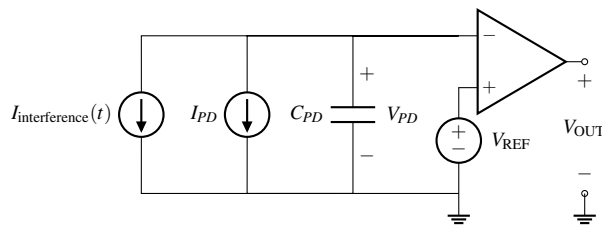


Figure 3: Model with interference

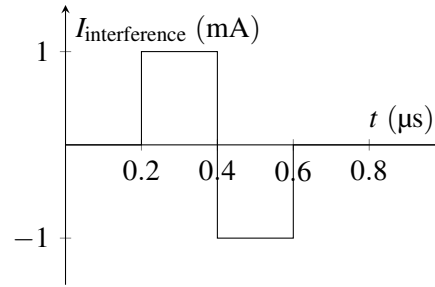


Figure 4: Interference current for part (c)

For this subpart, use the interference current defined in Figure 4 and the values in Table 1 (repeated for convenience).

Parameter	Unit	Value
I_{PD}	A	$1 \cdot 10^{-3}$
C_{PD}	F	$1 \cdot 10^{-9}$
V_{REF}	V	0.5
V_{DD}	V	1.0

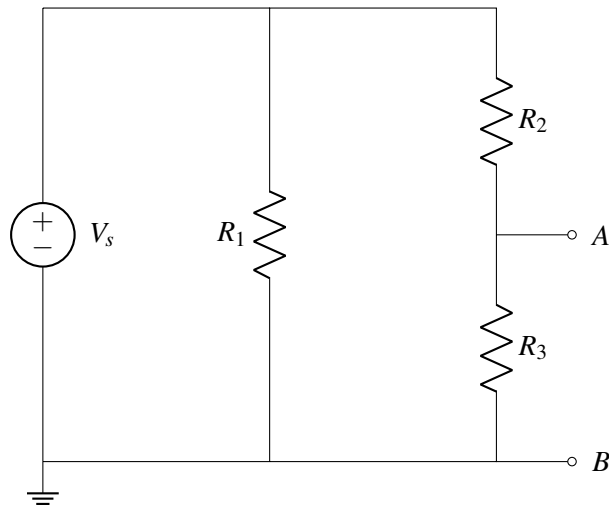
Table 1 repeated for convenience

Draw $V_{PD}(t)$ in the graph below given $V_{PD}(0) = V_{DD}$. Clearly label $\frac{dV_{PD}}{dt}$ for every line segment in your graph with numerical values. Also find t_{flip} where $V_{PD}(t_{\text{flip}}) = V_{REF}$.

Hint: $V_{PD}(t)$ is continuous.

2. Thévenin/Norton Equivalence

- (a) Find the Thévenin resistance R_{th} of the circuit shown below, with respect to its terminals A and B .



- (b) Now, a load resistor, $R_L = R$, is connected across terminals A and B , as shown in the circuit below. Find the power dissipated in the load resistor in terms of the given variables.

